YAKEEN

NEET 2.0 2027

PHYSICS | **Lecture-02**

Units and Measurement

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Topics to be covered

1. Physical Quantities || Dimension formula || Question Practice

195 → 645

185 → 630

200 marks → 600 marks

10+

1.1L → 700

1500

50+

80k → 700

70k → 1100

1L → 2200

1.8L → 2500

5000

Units and Measurements

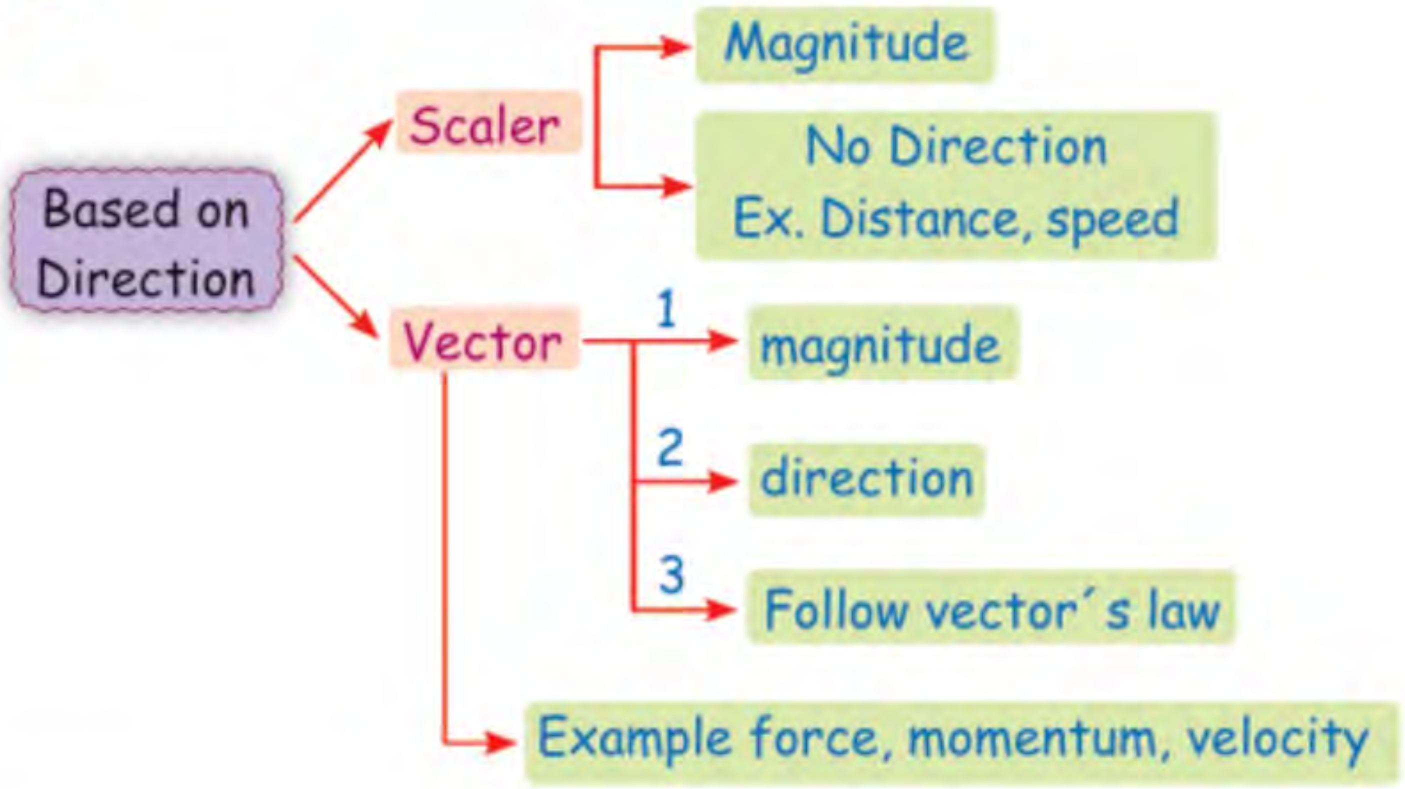
तो चलिए हमारी इस beautifull journey को start करते है पहले chapter unit and dimensions से.... ये chapter Advance/Mains/NEET तीनो के लिए important है..... मुझे पता है पहले chapter मे आपका excitation level बहुत high होता है.... from basic to advance का content हम इस chapter मे cover करेंगे मै बस इतना बोलुंगा हो सकता है शुरू मे ये chapter आपको थोड़ा मुश्किल लगे but at the end of physics ये chapter आपको सबसे आसान लगेगा.... from basic to Advance all relevent theory and question i will cover in this book.... तो चलिए शुरू करते है are you ready

PHYSICAL QUANTITIES

All quantities that can be measured are called physical quantities. e.g. time, length, mass, force, workdone, etc.

Classification of Physical Quantity

1



क्या होता है vector क्या होता है vector's law अभी हमे इन पर बिल्कुल ध्यान नही देना है ये चीजे हम vector वाले chapter मे detail मे पढ़ेंगे।

2. Based on dependency
   - Fundamental physical quantity
   - Derived physical quantity

Based on their Dependency

1. **Fundamental or base quantities:** A set of physical quantities which are completely independent of each other and all other physical quantities can be expressed in terms of these physical quantities is called Set of Fundamental Quantities.

2. **Derived quantities:** The quantities which can be expressed in terms of the fundamental quantities are known as derived quantities. Ex. Speed (= distance/time), volume, acceleration, force, pressure, etc.

Fundamental or base quantities

S. No.	Physical quantity	SI unit	Symbol
1.	Length	metre	m
2.	Mass	kilogram	kg
3.	Time	second	s
4.	Temperature	Kelvin	K
5.	Electric current	ampere	A
6.	Luminous intensity	candela	cd
7.	Amount of substance	mole	mol

A quantity can be expressed as numerical value and unit.

Quantity (Q) = $n_1U_1 = n_2U_2$

→ Physical quantity to measure.

$n_1 \rightarrow$ numerical value in 1^{st} system of measurement

$U_1 \rightarrow$ Units value in 1^{st} system of measurement

$n_2 \rightarrow$ numerical value in 2^{nd} system of measurement

$U_2 \rightarrow$ Units value in 2^{nd} system of measurement

If $U_1 > U_2 \Rightarrow n_1 < n_2$

If unit is bigger then numerical value is smaller and vice versa

Classification of Units

meter kilogram sec. MKS System	CGS System	FPS System
mass $\rightarrow$ kg	mass $\rightarrow$ gram	mass $\rightarrow$ pound
length $\rightarrow$ m	length $\rightarrow$ cm	length $\rightarrow$ foot
time $\rightarrow$ sec	time $\rightarrow$ sec	time $\rightarrow$ sec

Q. Convert Density = 50 kg/m^3 into CGS.

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Sol. Density $= 50\,kg/m^3 = \dfrac{50 \times 1000\,gm}{(100\,cm)^3} = .05\,gm/(cm)^3$

Q. Convert Density = 0.6 gm/cc into MKS.

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Sol. Density = 0.6 gm/cc $= \dfrac{0.6 \times \dfrac{kg}{1000}}{\left(\dfrac{1}{100} m\right)^3} = \dfrac{600\,kg}{m^3}$

DIMENSIONS

Dimensions of a physical quantity are the powers (or exponents) to which the base quantities are raised to represent that quantity.

The physical quantity that is expressed in terms of base quantities is enclosed in square brackets to remind that the equation is among the dimensions and not among the magnitude, such expression for physical quantity is called dimension formula.

meter वाला 'm' नही हैं

Mass has dimension	[M]
Length has dimension	[L]
Time has dimension	[T]
Temp. has dimension	[K]
Electric current	[A]

Q. Write dimensional formula of speed.

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Sol. $$\text{Speed} = \frac{\text{Distance}}{\text{Time}}$$

Dimensionally we can write

$$[\text{Speed}] = \left[\frac{\text{Distance}}{\text{Time}}\right] = \frac{[L]}{[T]} = [LT^{-1}]$$

So, dimensional formula of speed can be written as $LT^{-1} = [M^0 L^1 T^{-1}]$

Here

Dimension of mass $\rightarrow 0$

Dimension of length $\rightarrow 1$

Dimension of time $\rightarrow -1$

सर speed का Dimensional Formula (DF) क्या हम only LT^{-1} लिख सकते है क्या इसे box मे बंद करना जरूरी है

वैसे तो speed का DF हम LT^{-1}, $[LT^{-1}]$, $[M^0LT^{-1}]$ तीनो लिख सकते है लेकिन box लगाना अच्छी बात है, हम अकसर physical quantity को M, L, T मे represent करते है अगर तुम school मे exam दे रहे हो तो box जरूर लगाओ ताकी school वाले master sahab number ना काटे लेकिन अगर तुम JEE का test/mock test/question practice कर रहे हो तो box लगाने मे time waste मत करना।

Q. Find the dimensions of mass length & time in density.

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Sol. Formula of density is

$$[\text{Density}] = \frac{[\text{mass}]}{[\text{volume}]}$$

$$[\text{Density}] = \frac{M}{L \cdot L \cdot L} = \frac{M}{L^3} \Rightarrow \left[M^1 L^{-3} T^0\right]$$

Hence in density we can say

Dimensional of mass $\rightarrow 1$

Dimension of length $\rightarrow -3$

Dimension of time $\rightarrow 0$

1. **Fundamental or base units:**

   The units of fundamental quantities are called *base units*. In SI there are seven base units.

2. **Derived units:**

   The units of derived quantities or the units that can be expressed in terms of the base units are called <u>*derived*</u> units

   Ex. Unit of speed can be derived from units of distance & unit of time.

$$\text{Unit of speed} = \frac{\text{unit of distance}}{\text{unit of time}} = \frac{\text{metre}}{\text{second}} = ms^{-1}$$

Dimensions of Quantities Related to Mechanics

S.N.	Quantity	Formula	Unit	Dimension
1.	Velocity or speed (v)	$v = \frac{d}{t} = \frac{\text{Displacement or Distance}}{\text{Time}}$	m/s	$[M^0L^1T^{-1}]$
2.	Acceleration (a)	$a = \frac{\Delta v}{\Delta t} = \frac{\text{Change in velocity}}{\text{Change in time}}$	m/s^2	$[M^0LT^{-2}]$
3.	Momentum (P)	P = mv = Mass × Velocity	kg - m/s	$[M^1L^1T^{-1}]$
4.	Impulse (I)	$I = F \times \Delta t$ Force × Time	Newton-second or kg - m/s	$[M^1L^1T^{-1}]$
5.	Force (F)	F = ma = Mass × Acceleration	Newton	$[M^1L^1T^{-2}]$
6.	Pressure (P)	$P = \frac{F}{A} = \frac{\text{Force}}{\text{Area}}$	Pascal	$[M^1L^{-1}T^{-2}]$
7.	Kinetic energy (K_E)	$K = \frac{1}{2}mv^2 = \frac{1}{2}$ Mass × Velocity^2	Joule	$[M^1L^2T^{-2}]$

8.	Power (P)	$P = \frac{W}{t} = \frac{\text{Work}}{\text{Time}}$	Watt or Joule-sec	$[M^1L^2T^{-3}]$
9.	Density (d)	$\rho = \frac{m}{V} = \frac{\text{Mass}}{\text{Volume}}$	kg/m^3	$[M^1L^{-3}T^0]$
10.	Angular displacement (θ)	$\theta = \frac{S}{r} = \frac{\text{Arc}}{\text{Length}}$	Radian (rad)	$[M^0L^0T^0]$
11.	Angular velocity (ω)	$\omega = \frac{\Delta\theta}{\Delta t} = \frac{\text{Angular displacement}}{\text{Time}}$	Radian/sec	$[M^0L^0T^{-1}]$
12.	Angular acceleration (α)	$\alpha = \frac{\Delta\omega}{\Delta t} = \frac{\text{Angular velocity}}{\text{Time}}$	Radian/sec^2	$[M^0L^0T^{-2}]$
13.	Moment of inertia (I)	$I = mx^2 = \text{Mass} \times \text{Distance}^2$	$kg\text{-}m^2$	$[M^1L^2T^0]$

S.N.	Quantity	Formula	Unit	Dimension
14.	Torque (τ)	$\tau = F \times r_{\perp}$ = Force × Perpendicular distance	Newton-meter	$[M^1L^2T^{-2}]$
15.	Angular momentum (L)	L = mvr = Mass × Velocity × Radius	Joule-sec	$[M^1L^2T^{-1}]$
16.	Force constant or spring constant (k)	F = -kx = Force constant × Displacement	Newton/m	$[M^1L^0T^{-2}]$
17.	Gravitational constant (G)	$F = \frac{Gm_1m_2}{r^2}$ $= \frac{\text{Gravitational constant} \times \text{Mass}^2}{\text{Distance}^2}$	$N\text{-}m^2/kg^2$	$[M^{-1}L^3T^{-2}]$
18.	Gas constant (R)	PV = nRT Pressure = Velocity × Gas constant × Temperature	Joule/mol-K	$[M^1L^2T^{-2}\theta^{-1}]$
19.	Planck's constant (h)	E = hv Energy = Planck's constant × Frequency	Joule-s	$[M^1L^2T^{-1}]$
20.	Surface tension (S)	$S = \frac{F}{L} \Rightarrow \text{Surface Tension} = \frac{\text{Force}}{\text{Length}}$	N/m or Joule/m^2	$[M^1L^0T^{-2}]$

21.	Coefficient of viscosity (η)	$\eta = \frac{F}{6\pi ru} = \frac{\text{Force}}{\text{Radius} \times \text{Velocity}}$	kg/m-s	$[M^1L^{-1}T^{-1}]$
22.	Time period (T)	$T = \frac{1}{n} = \frac{1}{\text{Frequency}}$	Second	$[M^0L^0T^1]$
23.	Frequency (n)	$n = \frac{1}{T} = \frac{1}{\text{Time}}$	Hz	$[M^0L^0T^{-1}]$

Dimensions of Quantities Related to Heat, Thermodynamics and Kinetic Theory of Gases

S.N.	Quantity	Formula	Unit	Dimension
1.	Heat (Q)	Energy	Joule	$[ML^2T^{-2}]$
2.	Specific Heat (c)	$c = \frac{Q}{m \times \Delta\theta} = \frac{\text{Heat}}{\text{Mass} \times \text{Temperature}}$	Joule/kg-K	$[M^0L^2T^{-2}K^{-1}]$
3.	Thermal capacity (K)	$K = \frac{Q}{\Delta t} = \frac{\text{Heat}}{\text{Time}}$	Joule/K	$[M^1L^2T^{-2}K^{-1}]$
4.	Latent heat (L)	$L = \frac{Q}{m} = \frac{\text{Heat}}{\text{Mass}}$	Joule/kg	$[M^0L^2T^{-2}]$
5.	Boltzmann constant (k)	$k = \frac{2}{3}\frac{E}{T} = \frac{\text{Energy}}{\text{Temperature}}$	Joule/K	$[M^1L^2T^{-2}K^{-1}]$

S.N.	Quantity	Formula	Unit	Dimension
6.	Coefficient of thermal conductivity (k)	$k = \frac{Qd}{A \times \Delta\theta \times t} = \frac{\text{Heat} \times \text{Distance}}{\text{Area} \times \text{Temp. difference} \times \text{Time}}$	Joule/m-s-K	$[M^1L^1T^{-3}K^{-1}]$
7.	Stefan's constant (σ)	$\sigma = \frac{\Delta E}{A \times \Delta t \times \theta^4} = \frac{\text{Energy}}{\text{Area} \times \text{Time} \times \text{Temperature}^4}$	Watt/m^2-K^4	$[M^1L^0T^{-3}K^{-4}]$
8.	Wien's constant (b)	$b = \lambda_{max} \times T = \text{Wavelength} \times \text{Temperature}$	Meter-K	$[M^0L^1T^0K^1]$
9.	Coefficient of linear expansion (α)	$\alpha = \frac{\Delta L}{L}\frac{1}{T} = \frac{\text{Change in length}}{\text{Length} \times \text{Temperature}}$	Kelvin^{-1}	$[M^0L^0T^0K^{-1}]$
10.	Mechanical eq. of Heat (J)	$J = \frac{W}{Q} = \frac{\text{Work}}{\text{Heat}}$	Joule/Calorie	$[M^0L^0T^0]$
11.	Vander wall's constant (a)	$a = \frac{RTV^2}{V - b} - PV^2$	Newton-m^4	$[M^1L^5T^{-2}]$

12.	Vander wall's constant (b)	Same as Volume (V)	m^3	$[M^0L^3T^0]$
13.	Temperature (T)	$T = \frac{Q}{M\Delta C}$	Kelvin (K)	$[M^0L^0T^0K^1]$

Electricity & Magnetism

S.N.	Quantity	Formula	Unit	Dimension
1.	Electric charge (q)	q = I × t = Electric current × Time	Coulomb	$[M^0L^0T^1A^1]$
2.	Electric current (i)	$I = \frac{q}{t} = \frac{\text{Charge}}{\text{Time}}$	Ampere	$[M^0L^0T^0A^1]$
3.	Capacitance (C)	$C = \frac{q}{V} = \frac{\text{Charge}}{\text{Voltage difference}}$	Coulomb/volt or Farad	$[M^{-1}L^{-2}T^4A^2]$
4.	Electric potential (V)	$V = \frac{q}{C} = \frac{\text{Charge}}{\text{Capacitance}}$	Joule/coulamb	$[M^1L^2T^{-3}A^{-1}]$
5.	Permittivity of free space (ε)	$\frac{\text{Charge}^2}{4\pi \times \text{Electric force} \times \text{Distance}^2}$	$\frac{\text{Coulomb}^2}{\text{Newton} - \text{meter}^2}$	$[M^{-1}L^{-3}T^4A^2]$
6.	Dielectric constant (K)	$K = \frac{\varepsilon}{\varepsilon_0} = \frac{\text{Permittivity in medium}}{\text{Permittivity in free space}}$	Unitless	$[M^0L^0T^0]$

S.N.	Quantity	Formula	Unit	Dimension
7.	Resistance (R)	$R = \frac{V}{I} = \frac{\text{Voltage difference}}{\text{Electric current}}$	Volt/Ampere or Ohm	$[M^1L^2T^{-3}A^{-2}]$
8.	Resistivity or Specific resistance (ρ)	$\rho = \frac{RA}{L} = \frac{\text{Resistance} \times \text{Area}}{\text{Length}}$	Ohm-meter	$[M^1L^3T^{-3}A^{-2}]$
9.	Coefficient of self-induction (L)	$L = \frac{\mu_0 N^2 A}{I}$	$\frac{\text{Volt} - \text{Second}}{\text{Ampere}}$ or Henery Ohm-second	$[M^1L^2T^{-2}A^{-2}]$
10.	Magnetic flux (ϕ)	$\phi = B \times A$ = Magnetic field × Area	Volt-second or Weber	$[M^1L^2T^{-2}A^{-1}]$
11.	Magnetic induction (B)	$B = \frac{F}{q \times v} = \frac{\text{Magnetic force}}{\text{Charge} \times \text{Velocity}}$	$\frac{\text{Newton}}{\text{Ampere} - \text{meter}}$ or Tesla	$[M^1L^0T^{-2}A^{-1}]$
12.	Magnetic Intensity (H)	$H = \frac{B}{\mu} = \frac{\text{Magnetic field}}{\text{Permeability}}$	Ampere/meter	$[M^0L^{-1}T^0A^1]$

13.	Magnetic dipole moment (M)	$M = I \times A$ = Current × Area	Ampere-meter^2	$[M^0L^2T^0A^1]$
14.	Permeability of free space (μ_0)	$\mu_0 = \frac{B \cdot \ell}{I} = \frac{\text{Magnetic field} \times \text{Length}}{\text{Current}}$	$\frac{\text{Newton}}{\text{Ampere}^2}$	$[M^1L^1T^{-2}A^{-2}]$
15.	Surface charge density (σ)	$\sigma = \frac{q}{A} = \frac{\text{Charge}}{\text{Area}}$	Coulomb-meter^2	$[M^0L^{-2}T^0A^1]$
16.	Electric dipole moment (p)	$p = q \times d$ = Charge × Distance	Coulomb-meter	$[M^0L^1T^1A^1]$
17.	Conductance (G)	$G = \frac{1}{R} = \frac{1}{\text{Resistance}}$	Ohm^{-1}	$[M^{-1}L^{-2}T^3A^1]$
18.	Conductivity (σ)	$\sigma = \frac{1}{\rho} = \frac{1}{\text{Resistivity}}$	Ohm^{-1} meter^1	$[M^{-1}L^{-3}T^3A^2]$
19.	Current density (J)	$J = \frac{I}{A} = \frac{\text{Current}}{\text{Area}}$	Ampere/m^2	$[M^0L^{-2}T^0A^1]$
20.	Intensity of electric field (E)	$E = \frac{F}{q} = \frac{\text{Electric force}}{\text{Electric charge}}$	Volt/meter, Neuton/coulomb	$[M^1L^1T^{-3}A^{-1}]$
21.	Rydberg constant (R)	$R_H = \frac{me^4}{8h^3c\varepsilon_0^2}$	m^{-1}	$[M^0L^{-1}T^0]$

Quantities Having Same Dimensions

S.N.	Dimension	Quantity
1.	$[M^0L^0T^{-1}]$	Frequency, Angular frequency, Angular velocity and Velocity gradient
2.	$[M^1L^2T^{-2}]$	Work, Internal energy, Potential energy, Kinetic energy. Torque
3.	$[M^1L^{-1}T^{-2}]$	Pressure, Stress, Young's modulus, Bulk modulus, Modulus of rigidity, Energy density
4.	$[M^1L^1T^{-1}]$	Momentum, Impulse
5.	$[M^1L^1T^{-2}]$	Thrust, Force, Weight
6.	$[M^1L^2T^{-1}]$	Angular momentum and Planck's constant
7.	$[M^1L^0T^{-2}]$	Surface tension, Surface energy (energy per unit area), Force constant and Spring constant
8.	$[M^0L^2T^{-2}]$	Latent heat and Gravitational potential
9.	$[M^1L^2T^{-2}\theta^{-1}]$	Thermal capacity, Gas constant and Entropy
10.	$[M^0L^0T^1]$	L/R, $\sqrt{LC}$, RC where L = Inductance, R = Resistance, C = Capacitance

11.	$[M^0L^1T^0]$	Distance, Displacement, Radius, Wavelength, radius of gyration.
12.	$[M^0L^1T^{-1}]$	Speed, Velocity, Velocity of light.
13.	$[M^0L^1T^{-2}]$	Acceleration, Acceleration due to gravity, Centripetal acceleration.
14.	$[M^0L^0T^{-1}]$	Decay constant, Rate of disintegration.

Units of Some Physical Quantities in Different Systems

Type of Physical Quantity	Physical Quantity	CGS (Originated in France)	MKS (Originated in France)	FPS (Originated in Britain)
Fundamental	Length	cm	m	ft
	Mass	g	kg	lb
	Time	s	s	s
Derived	Force	dyne	newton(N)	poundal
	Work or Energy	erg	joule(J)	ft-poundal
	Power	erg/s	watt(W)	ft-poundal/s

Units and Dimensions of Physical Quantities

Quantity	Common Symbol	SI unit	Dimension
Displacement	s	METRE (m)	L
Mass	m, M	KILOGRAM (kg)	M
Time	t	SECOND (s)	T
Area	A	m^2	L^2
Volume	V	m^3	L^3
Density	ρ	kg/m^3	M/L^3
Velocity	v, u	m/s	L/T
Acceleration	a	m/s^2	L/T^2
Force	F	newton (N)	ML/T^2
Work	W	joule (J) (= N-m)	ML^2/T^2
Energy	E, U, K	joule (J)	ML^2/T^2
Power	P	watt (W) (= J/s)	ML^2/T^3
Momentum	p	kg-m/s	ML/T
Gravitational constant	G	$N\text{-}m^2/kg^2$	L^3/MT^2

Angle	θ, φ	radian	
Angular velocity	ω	radian/s	T^{-1}
Angular acceleration	α	radian/s^2	T^{-2}
Angular momentum	L	Kg-m^2/s	ML^2/T
Moment of inertia	I	Kg-m^2	ML^2
Torque	τ	N-m	ML^2/T^2
Angular frequency	ω	radian/s	T^{-1}
Frequency	υ	hertz (Hz)	T^{-1}
Period	T	s	T
Young's modulus	Y	N/m^2	M/LT^2
Bulk modulus	B	N/m^2	M/LT^2
Shear modulus	η	N/m^2	ML/T^2
Surface tension	S	N/m	M/T^2
Coefficient of viscosity	η	N-s/m^2	M/LT

Pressure	P, p	N/m^2, Pa	M/LT^2
Wavelength	λ	m	L
Intensity of wave	I	W/m^2	M/T^3
Temperature	T	KELVIN (K)	K
Specific heat capacity	c	J/kg-K	L^2/T^2K
Stefan's constant	σ	W/m^2- K^4	M/T^3K^4
Heat	Q	J	ML^2/T^2
Thermal conductivity	K	W/m-K	ML/T^3K

सच-सच बताओ क्या तुम हर physical quantity का unit और dimensional formula रट रहे हो ना

Haa humne RATT liye

Nhi RATTNA tha

पूरे दो साल यही पढ़ना है कि कौन-सी physical quantity क्या है So, इस पर time बिलकुल waste ना करें और मस्त रहे अगर याद करना ही है तो velocity momentum, acc, force, work, power, kinetic energy याद कर सकते हो

QUESTION

$\frac{MLT^{-2}}{L}$

H/W

Match the list-I with List-II.

Choose the correct answer from the options given below: [30 January 2024 - Shift 1]

	List-I		List-II
A.	Coefficient of viscosity	I.	$[ML^2T^{-2}]$
B.	Surface Tension $= \frac{F}{l}$	II.	$[ML^2T^{-1}]$
C.	Angular momentum	III.	$[ML^{-1}T^{-1}]$
D.	Rotational kinetic energy	IV.	$[ML^0T^{-2}]$

1. A – II, B – I, C – IV, D – III
2. A – I, B – II, C – III, D – IV
3. A – III, B – IV, C – II, D – I
4. A – IV, B – III, C – II, D – I

Ans. (3)

QUESTION

The dimensional formula of $\boxed{\text{angular impulse}}$ is: [01 February 2024 - Shift 1]

$\tau \times$ time

$ML^2T^{-2} \cdot T$

1. $[ML^{-2}T^{-1}]$
2. $[ML^{2}T^{-2}]$
3. $[MLT^{-1}]$
4. $[ML^{2}T^{-1}]$

Ans. (4)

QUESTION

H/w

Match List I with List II and select the correct answer using the codes given below the lists:

[Adv. 2013]

List I	List II
P. Boltzmann constant	1. $[ML^2T^{-1}]$
Q. Coefficient of viscosity	2. $[ML^{-1}T^{-1}]$
R. Planck constant	3. $[MLT^{-3}K^{-1}]$
S. Thermal conductivity	4. $[M^2L^{-2}K^{-1}]$

	P	Q	R	S
(1)	3	1	2	4
(2)	3	2	1	4
(3)	4	2	1	3
(4)	4	1	2	3

Ans: (3)

Q $\dfrac{\Delta Q}{\Delta t} = \dfrac{KA\,\Delta T}{l}$.

$K \Rightarrow \dfrac{\Delta Q}{\Delta t}\,\dfrac{l}{A\,\Delta T} \Rightarrow \dfrac{ML^2T^{-2}\cdot L}{T\cdot L^2\cdot K}$

$\Delta Q \rightarrow$ heat

$l \rightarrow$ length

$\Delta T \rightarrow$ change in temp.

$A \rightarrow$ Area

$K \rightarrow$ coff. of thermal conductivity.

QUESTION - 02

The damping force on an oscillator is directly proportional to the velocity. The units of the constant of proportionality are: [2012]

(1) kg m s^{-1} (2) kg m s^{-2}
(3) kg s^{-1} (4) kg s

$F \propto v$

$F = kv$

$k \Rightarrow \frac{F}{v} \Rightarrow \frac{MLT^{-2}}{LT^{-1}} = MT^{-1}$

$k \Rightarrow \frac{F}{v} \Rightarrow kg\,(sec^{-1})$

Ans : (3)

QUESTION - 14

Plane angle and solid angle have:

[2022]

(1) Units but no dimensions

(2) Dimensions but no units

(3) No units and no dimensions

(4) Both units and dimension

Ans : (1)

QUESTION - 15

If E and G respectively denote energy and gravitational constant, then E/G has the dimensions of:

[2021]

(1) $[M^2][L^{-2}][T^{-1}]$ (2) $[M^2][L^{-1}][T^0]$
(3) $[M][L^{-1}][T^{-1}]$ (4) $[M][L^0][T^0]$

$$\frac{E}{G} = \frac{ML^2T^{-2}}{M^{-1}L^3T^{-2}} = M^2L^{-1}$$

Ans : (2)

QUESTION - 16

Dimensions of stress are:

[2020]

(1) $[MLT^{-2}]$ (b) $[ML^{2}T^{-2}]$

(3) $[ML^{0}T^{-2}]$ (4) $[ML^{-1}T^{-2}]$

$$\frac{F}{A} = \frac{MLT^{-2}}{L^{2}}$$

Ans : (4)

QUESTION - 17

The pair of quantities having same dimensions is:

[Karnataka NEET 2013]

(1) Impulse and Surface Tension

(2) Angular momentum and Work

(3) Work and Torque

(4) Young's modulus and Energy

Ans : (3)

Q $\frac{F}{A} = Y\left(\frac{\Delta l}{l}\right)$

$Y \Rightarrow \frac{F}{A} \Rightarrow \frac{MLT^{-2}}{L^2} \Rightarrow M L^{-1} T^{-2}$

$F \rightarrow$ Force
$A \rightarrow$ Area
$\Delta l \longrightarrow$ change in length
$l \longrightarrow$ length

QUESTION - 24

The dimensions of universal gravitational constant are:

[2004, 1992]

(1) $[M^{-1}L^{3}T^{-2}]$ (2) $[ML^{2}T^{-1}]$

(3) $[M^{-2}L^{3}T^{-2}]$ (4) $[M^{-2}L^{2}T^{-1}]$

Ans : (1)

QUESTION - 25

The dimensions of Planck's constant equals to that of

[2001]

(1) energy

(2) momentum

(3) angular momentum

(4) power

HW

Ans : (3)

QUESTION - 26

Which pair do not have equal dimensions?

[2000]

(1) Energy and torque

(2) Force and impulse

(3) Angular momentum and Planck's constant

(4) Elastic modulus and pressure

Ans : (2)

(D.F.)

① Distance $\longrightarrow$ L

② Radius $\longrightarrow$ L

③ Displacement $\longrightarrow$ L

④ Speed = $\frac{\text{Distance}}{\text{time}}$ $\longrightarrow$ $\frac{L}{T} = LT^{-1}$

⑤ Velocity = $\frac{\text{Displacement}}{\text{time}}$ $\longrightarrow$ $\frac{L}{T} = LT^{-1}$

⑥ Acc. = $\frac{\text{Velocity}}{\text{time}}$ $\longrightarrow$ $\frac{LT^{-1}}{T} = LT^{-2}$

⑦ Acc. due to gravity (g) $\Rightarrow LT^{-2}$

⑧ Force (F = mass x acc) $\Rightarrow [MLT^{-2}]$

⑨ work = (Force x Distance)

$\Rightarrow [MLT^{-2} \times L]$

$= [ML^2T^{-2}]$

⑩ Power = $\frac{\text{work}}{\text{time}} \Rightarrow \frac{ML^2T^{-2}}{T}$

$= [ML^2T^{-3}]$

(11) Potential energy $\Rightarrow (U = mgh) \Longrightarrow M LT^{-2} \cdot L = [ML^2T^{-2}]$

(12) kinetic energy $\left(K.E. = \frac{1}{2}mv^2\right) \Longrightarrow M(LT^{-1})^2 = [ML^2T^{-2}]$

→ same D.F

(13) Torque $= (\text{Force} \times \text{Distance}) \Rightarrow MLT^{-2} \cdot L = [ML^2T^{-2}]$

(14) Pressure $= \left(\frac{\text{Force}}{\text{Area}}\right) \Rightarrow \frac{MLT^{-2}}{L^2} = ML^{-1}T^{-2}$

(15) Stress $= \frac{\text{Force}}{\text{Area}} \Rightarrow \frac{MLT^{-2}}{L^2} = ML^{-1}T^{-2}$

(16) Surface tension $= \frac{\text{Force}}{\text{length}} \Rightarrow \frac{MLT^{-2}}{L} = ML^0T^{-2}$

(17) Angle $= \dfrac{\text{Arc length}}{\text{Radius}} = \dfrac{L}{L} = 1 = M^0L^0T^0$

(Dimensionless) (Unit → radian)

(18) freq. $= \dfrac{1}{\text{time period}} \Rightarrow \dfrac{1}{T} = T^{-1}$

(19) Angular velocity $= \left(\dfrac{\text{Angle}}{\text{time}}\right) \Rightarrow \dfrac{1}{T} = T^{-1}$

(20) Angular Acc. $\left(\dfrac{\text{Angular velocity}}{\text{time}}\right) \Rightarrow \dfrac{T^{-1}}{T}$

$= T^{-2}$

(21) momentum, $(P = mv)$

$\Rightarrow MLT^{-1}$

(22) Angular momentum $= mvr$

(mass × velocity × Distance)

$\Rightarrow MLT^{-1} \cdot L = ML^2T^{-1}$

(23) moment of inertia $(I = MR^2)$

$\Rightarrow ML^2$

Distance

Spring pot. energy

Nuclear Energy

Binding energy

Kinetic Energy

Internal Energy

Potential Energy

ML^2T^{-2}

work

Surface energy

magnetic Energy

Thermal Energy

threshold energy

Energy of one photon

heat energy

Imp.

velocity $\longrightarrow LT^{-1}$

Acc $\longrightarrow LT^{-2}$

Force $\longrightarrow MLT^{-2}$

Energy $\longrightarrow ML^{2}T^{-2}$

Centripital force

psedo force

Normal Force

Viscous force

Force $\rightarrow MLT^{-2}$

friction force

Buoyant force

Gravitational force

Spring force

tension force

Nuclear force

magnetic force

Electrost. force

Lorentz force

Dhokebaj

Electro motive force.

Emf $= MLT^{-2}$ X

$10V = \text{Emf}$

$$\text{potential} = \frac{P.E.}{\text{Charge}} = \frac{ML^2T^{-2}}{AT}$$

Drift velocity

wave velocity

orbital velocity

Velocity $\rightarrow LT^{-1}$

Relative velocity

Terminal Velocity

escape Velocity

Rms speed

Dhokebaj

Angular velocity $= LT^{-1}$ X

$= T^{-1}$

(24) Impulse = (Force × time) $\Rightarrow MLT^{-2} \cdot T = MLT^{-1}$

(25) Angular Impulse = (Torque × time) $\Rightarrow ML^2T^{-2} \cdot T = ML^2T^{-1}$

(26) Power = $\left(\frac{\text{Energy}}{\text{time}}\right) \Rightarrow \frac{ML^2T^{-2}}{T} = ML^2T^{-3}$

(27) Intensity = $\left(\frac{\text{Power}}{\text{Area}}\right) \Rightarrow \frac{ML^2T^{-3}}{L^2} = MT^{-3}$

Intensity = $\frac{\text{Energy}}{\text{Area time}} \Rightarrow \frac{ML^2T^{-2}}{L^2 \cdot T} = MT^{-3}$

(28) Temp. Gradient $\Rightarrow \frac{K}{L} \Rightarrow M^0 L^{-1} T^0 K^1$

(29) Velocity Gradient $\Rightarrow \frac{LT^{-1}}{L} \Rightarrow T^{-1}$

(30) Pressure Gradient $\Rightarrow \frac{MLT^{-2}}{L^2 \cdot L} = M L^{-2} T^{-2}$

(31) Change in Length $\Rightarrow L$

(32) Change in time $\Rightarrow T$

Change in Velocity $\Rightarrow LT^{-1}$

Change in length $= L_f - L_i$

$\Longrightarrow$ D.F $\Rightarrow L$

(33) $\Delta l \longrightarrow L$

(34) ΔP = change in momentum

(D.F) $\Rightarrow$ momentum $\Rightarrow MLT^{-1}$

(35) $\Delta \vec{V}$ = Change in Velocity $\Rightarrow LT^{-1}$

(36) $d\vec{v} \longrightarrow LT^{-1}$

Very very small Change in Velocity

velocity $v \longrightarrow LT^{-1}$

$\Delta V \longrightarrow LT^{-1}$

Change in velocity $\longrightarrow LT^{-1}$

$d\vec{v} \longrightarrow LT^{-1}$

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- $d\vec{F} \Rightarrow MLT^{-2}$
- $dt \Rightarrow T$
- $\dfrac{dF}{dt} = \dfrac{F}{t} \Rightarrow \dfrac{MLT^{-2}}{T}$
- $\dfrac{\Delta F}{\Delta t} = \dfrac{F}{t} \Rightarrow \dfrac{MLT^{-2}}{T}$

(38) Strain $= \frac{\Delta l}{l} \Rightarrow \frac{L}{L} = 1$ (Dimensionless)

(39) $\left.\begin{array}{l}\text{Solid angle}\\ \text{plane angle}\end{array}\right\} \Rightarrow$ Dimensionless.

(40) Force const $= \left(\frac{\text{Force}}{\text{length}}\right) \Rightarrow \frac{MLT^{-2}}{L} = MT^{-2}$

(41) find D.F. of G (Universal Grav. Const)

$$F = \frac{G m_1 m_2}{r^2}$$

$$\left\{ \begin{array}{l} F \longrightarrow \text{Force} \\ m_1, m_2 \longrightarrow \text{mass} \\ r \longrightarrow \text{Distance} \end{array} \right\}$$

Sol^n $G = \dfrac{F r^2}{m_1 m_2}$

$$G \Rightarrow \frac{MLT^{-2} \cdot L^2}{M\,M}$$

$$\boxed{G = M^{-1} L^3 T^{-2}}$$

$$G = \frac{F r^2}{m_1 m_2} \Rightarrow \frac{Nm^2}{(Kg)^2}$$

Unit

(5) If D.F. of Universal Grav. Const is given by $M^x L^y T^z \equiv M^{-1} L^3 T^{-2}$ then find (1) $x+y+z$.

$-1+3-2 = 0$

(2) $|x| + |y| + |z|$

$|-1| + |3| + |-2| = 1+3+2$

$= 6$

(42) $E = h\nu$

$E \rightarrow$ Energy

$h \rightarrow$ planc const

$\nu \rightarrow$ freq.

Sol: $h = \frac{E}{\nu} \Rightarrow \frac{ML^2T^{-2}}{T^{-1}}$

$$\boxed{h \Rightarrow ML^2T^{-1}}$$

(43) $\eta \rightarrow$ coff of viscosity.

$F = 6\pi r \eta v$

$F \rightarrow$ force

$r \rightarrow$ Radius

$v \rightarrow$ Velocity.

Sol: $\eta = \frac{F}{6\pi r v}$

$\eta \Rightarrow \frac{MLT^{-2}}{L\,LT^{-1}} = ML^{-1}T^{-1}$

D.F. of specific heat const 'S'

(44) $\Delta Q = m S \Delta T$

$\Delta Q \longrightarrow$ heat energy

$m \longrightarrow$ mass

$S \longrightarrow$ specific heat const

$\Delta T \longrightarrow$ change in temp.

Sol

$$S = \frac{\Delta Q}{m \Delta T} \Rightarrow \frac{ML^2T^{-2}}{M\ K}$$

$$\Rightarrow M^0L^2T^{-2}K^{-1}$$

Find D.F. of latent heat.

(45) $\Delta Q = mL$

ΔQ → heat energy, m → mass, L → Latent heat

Sol

$$L = \frac{\Delta Q}{m} \Rightarrow \frac{ML^2T^{-2}}{M} = L^2T^{-2}$$

(46) Bolzman Const (K)

Energy ← $U = \frac{3}{2}KT$ → temp

① KT, $K\theta$ $\Rightarrow$ mL^2T^{-2} = (Boltzman const) × (temp)

temp

$$\boxed{K = \frac{2}{3}\frac{U}{T} \Rightarrow \frac{mL^2T^{-2}}{K} \Rightarrow mL^2T^{-2}K^{-1}}$$

Bolzman Const

(47) $PV = nRT$

$\text{pressure} = \frac{\text{Force}}{\text{Area}}$

$V \longrightarrow$ Volume

$n \longrightarrow$ no. of moles

$T \longrightarrow$ temp.

D.F. of $R \Rightarrow \frac{PV}{nT} \Rightarrow \frac{\frac{MLT^{-2}}{L^2} \cdot L^3}{\text{mol} \cdot K} \Rightarrow ML^2T^{-2}\text{mol}^{-1}K^{-1}$

48 Energy density $= \dfrac{\text{Energy}}{\text{Vol}} = \dfrac{ML^2T^{-2}}{L^3} = ML^{-1}T^{-2}$

$$\left(\frac{1}{2}\epsilon_0 E^2,\ \frac{1}{2}\frac{B^2}{\mu_0}\right)$$

SKC

* Jaruri Nahi hai ki sare Const dimensionless ho.

12th वाली phy. quantity

① Charge = (Current × time) ⇒ AT

$$\boxed{\text{Charge} \equiv AT}$$

② $F = qE$

force

Electric field

$$E \Rightarrow \frac{F}{q} \Rightarrow \frac{MLT^{-2}}{AT}$$

③ Electric Potential (V) = $\dfrac{\text{Potential energy}}{\text{Charge}}$

$$\frac{ML^2T^{-2}}{AT} = ML^2T^{-3}A^{-1}$$

④ $V = iR$

Resistance

$$R = \frac{V}{i} \Rightarrow \frac{ML^2T^{-3}A^{-1}}{A} \equiv ML^2T^{-3}A^{-2}$$

⑤ $R = \rho \dfrac{l}{A}$

Resistivity

A → Area

l → length

$$\rho = \frac{RA}{l} = \frac{(\ )\cdot L^2}{L}$$

$$\rho = \frac{1}{\sigma}$$

conductivity

⑥ $F = \frac{1}{4\pi\epsilon_0}\frac{q_1 q_2}{r^2}$

$\epsilon_0 \rightarrow$ permittivity of free space.

$\epsilon_0 \Rightarrow \frac{q_1 q_2}{4\pi F r^2} \Rightarrow \frac{AT\ AT}{MLT^{-2}L^2} = \checkmark$

⑦ $F = qvB$

force

charge

magnetic fi

velocity

$B = \frac{F}{qv} \Rightarrow \frac{MLT^{-2}}{AT \cdot LT^{-1}}$

⑧ $\phi = E \cdot A \cdot \Rightarrow \frac{MLT^{-2}}{AT} \cdot L^2$

Electric flux

E.F.

area

⑨ magnetic flux

$\phi = BA \Rightarrow (\checkmark)L^2$

magnetic field

Area

Q $B = \frac{\mu_0 i}{2R}$

$i \longrightarrow$ current

$R \longrightarrow$ Radius.

$\mu_0 = ?$

$$\boxed{B = \frac{F}{qv}} \Rightarrow B = \frac{MLT^{-2}}{AT\,LT^{-1}}$$

Sol^n

$$\mu_0 \Rightarrow \frac{2BR}{i} \Rightarrow \frac{MLT^{-2}}{AT\,LT^{-1}} \times \frac{L}{A} = \checkmark$$

Q $q = CV$

$q \to$ charge, $C \to$ (capacitance), $V \to$ potential

$$C = \frac{q}{V} \Rightarrow \frac{AT}{\left(\frac{ML^2T^{-2}}{AT}\right)} = \checkmark$$

Q $\phi = L\,i$

magnetic flux

current

Coff. of self induct.

Sol

$L = \frac{\phi}{i} \Rightarrow \frac{B \cdot Area}{i} = \frac{MLT^{-2}}{AT\,LT^{-1}} \frac{L^2}{A}$

= ✓

$F = qvB$

$B = \frac{F}{qv}$

Q

$$\mu = \frac{\text{speed of light in vaccum}}{\text{" " " medium}} \Rightarrow \frac{LT^{-1}}{LT^{-1}} \equiv 1$$

Refraction Index

Q

$$\frac{\Delta\theta}{\Delta t} = A\sigma T^4$$

σ → stefan const

Sol

$$\sigma = \frac{\Delta\theta}{\Delta t \cdot A\, T^4} \Rightarrow \frac{mL^2T^{-2}}{T \cdot L^2 K^4}$$

$\Delta\theta \longrightarrow$ heat

$A \longrightarrow$ Area

$T \longrightarrow$ temp.

सर speed का Dimensional Formula (DF) क्या हम only LT^{-1} लिख सकते है क्या इसे box मे बंद करना जरूरी है

वैसे तो speed का DF हम LT^{-1}, $[LT^{-1}]$, $[M^0LT^{-1}]$ तीनो लिख सकते है लेकिन box लगाना अच्छी बात है, हम अकसर physical quantity को M, L, T मे represent करते है अगर तुम school मे exam दे रहे हो तो box जरूर लगाओ ताकी school वाले master sahab number ना काटे लेकिन अगर तुम JEE का test/mock test/question practice कर रहे हो तो box लगाने मे time waste मत करना।

सच-सच बताओ क्या तुम हर physical quantity का unit और dimensional formula रट रहे हो ना

Haa humne RATT liye

Nhi RATTNA tha

पूरे दो साल यही पढ़ना है कि कौन-सी physical quantity क्या है So, इस पर time बिलकुल waste ना करें और मस्त रहे अगर याद करना ही है तो velocity momentum, acc, force, work, power, kinetic energy याद कर सकते हो

Home Work

- solve all phy. quantity of today class on rough copy

Thank

You